

Jacob Mattie

Vancouver
778-710-7554
j_mattie@live.ca
<https://github.com/qimbet>

Summary

Quantitative Engineer with a mathematics foundation and hands-on experience building scalable data systems, statistical models, and backtesting frameworks. Proactive and self-directed, with a strong grasp of financial concepts and a deep commitment to continuous study and growth in systematic investing.

Software: C++, Python, SQL, Redis, Selenium, Linux, Bash, Git

Key Skills: Nonlinear Optimization, Statistics, Linear Algebra

Work Experience

Investment Analyst

Aligned Capital Partners / River Wealth

Vancouver, BC

11/2024 - Ongoing

- Developed scheduled **web scraping and ETL pipelines** with Celery Beat and Redis, centralizing a custom PostgreSQL database from disparate public and proprietary sources.
- Founded the company's **data assets program** by engineering a time-stamped repository of key performance indicators (KPIs) across 5000+ securities, enabling time-series analysis and quantitative modeling.
- Configured a **Linux-based server stack** (Fedora, Gunicorn, Nginx) to host a Django-driven platform for surfacing financial KPIs, model outputs, and automated reporting pipelines to **cross-functional teams**.

Test Technologist

IBC Technologies

Burnaby, BC

6/2022 – 9/2023

- Digitized hardware testing by linking C++-based microcontrollers into **cloud-connected data pipelines**, enabling remote diagnostics and control for high-fidelity process validation.
- Automated analysis of high-volume datasets (40,000+ lines per series) using MATLAB, reducing processing time from 20 minutes to under 3 minutes per iteration, enabling rapid, reproducible **insight generation**.

Opinions Editor

The Peak Publications Society

Burnaby, BC

10/2020 – 6/2022

- Managed the Opinions Section of a weekly newspaper, **coordinating a rotating team** of contributors and production staff to publish 2500+ words across 5 articles while meeting strict production deadlines.
- **Mentored and coached** writers of varying experience levels, providing editorial feedback, thematic guidance, and structural training to develop their voice and align their work with the section's goals.

IT Technician (Internship)

BC Cancer Research Centre

Vancouver, BC

09/2018 – 01/2019

- Diagnosed and resolved complex technical issues across Windows, Mac, and Linux environments in a large-scale research facility, showing adaptability and **rapid problem-solving** skills with diverse technology stacks.
 - Configured and maintained network infrastructure and digital policy to ensure **cybersecurity** compliance and reliable data transmission across departments.
 - Created foundational documentation on daily operations and training methods, closing knowledge gaps and **standardizing processes** across future hires.
-

Projects

Hydroponics

Independent Project

04/2025 – 08/2025

- Engineered a fully autonomous hydroponics system able to grow 36 spinach plants from seed to harvest with no manual input.
 - Developed a **C++-based** control board to ingest real-time sensor data and execute dynamic response loops, stabilizing optimal growth conditions.
-

Education

Simon Fraser University

B.A.Sc in Mathematics

01/2019 - 05/2022

Certifications

Nuclear Energy Worker

Canadian Nuclear Safety Commission

09/2025 - 09/2028

AI and Technology Foundations

Canada's Ocean Supercluster

04/2025 - 06/2025

Volunteering & Leadership

Triumf - ASPIRE Astrochemistry Lab

Automation Technologist

Vancouver, British Columbia

05/2025 - Ongoing

- Developed a Bash-based installer for EPICS, a **data acquisition** and control system, leading a **modularization initiative** on multi-million dollar physics experiments.
- Integrated a Python motor controller into the laboratory architecture, **reducing experimental variance to 0%** and eliminating need for active engagement with the experiment over its hours-long runtime.